Natural Language Processing Overview

What Is Natural Language Processing?

Natural language processing, commonly abbreviated NLP, is the subfield of artificial intelligence concerned with enabling computers to understand, interpret, generate, and interact using human language, whether written text or spoken speech. Human language is notoriously difficult for computers to handle because it is ambiguous, highly context-dependent, and full of idioms, sarcasm, and implicit meaning that people navigate effortlessly but which do not have a single, precise mathematical definition. NLP sits at the intersection of computer science, linguistics, and machine learning, and it underlies many everyday technologies, including search engines, spam filters, machine translation services, voice assistants, and chatbots.

Classic NLP Tasks

Several tasks have become standard building blocks of the field. Tokenization is the process of splitting raw text into smaller units, called tokens, which are typically words, subwords, or punctuation marks; nearly every NLP pipeline begins with some form of tokenization. Part-of-speech tagging assigns a grammatical category, such as noun, verb, or adjective, to each token in a sentence. Named entity recognition identifies and classifies proper names in text, such as people, organizations, and locations. Sentiment analysis determines the emotional tone of a piece of text, for example whether a product review is positive, negative, or neutral, and is widely used by businesses to monitor customer feedback. Machine translation converts text from one language to another, and has improved dramatically since the introduction of neural network based approaches. Text summarization condenses a longer document into a shorter version that preserves its key information, either by selecting important sentences directly from the source, known as extractive summarization, or by generating new sentences that paraphrase the content, known as abstractive summarization. Question answering systems, which are directly relevant to retrieval-augmented generation systems, take a natural-language question and attempt to return a precise answer, often by locating relevant information within a document or corpus.

From Rule-Based Systems to Statistical NLP

Early NLP systems, built in the 1950s through the 1980s, relied heavily on hand-crafted grammatical rules and dictionaries designed by linguists. While interpretable, these rule-based systems were brittle and did not generalize well to the enormous variability of real human language. Starting in the late 1980s and accelerating through the 1990s and 2000s, the field shifted toward statistical NLP, in which probabilistic models were trained on large collections of text, called corpora, to learn patterns of language use automatically rather than relying on

hand-written rules. Techniques from this era include n-gram language models, which estimate the probability of a word based on the small number of words that preceded it, and hidden Markov models, which were widely used for tasks such as part-of-speech tagging and speech recognition.

Word Representations and Embeddings

A central challenge in applying machine learning to text is that computers operate on numbers, not words, so text must first be converted into a numerical representation. Early approaches, such as one-hot encoding and bag-of-words representations, treated each word as an independent symbol and ignored any notion of meaning or similarity between words. A major breakthrough came with word embeddings, popularized by methods such as Word2Vec and GloVe in the early 2010s, which represent each word as a dense vector of real numbers, typically a few hundred dimensions, learned in such a way that words with similar meanings end up close together in the vector space. For example, the vectors for "king" and "queen" end up near each other, and interestingly, arithmetic on these vectors can capture certain relationships, such as the vector for "king" minus "man" plus "woman" landing close to the vector for "queen." These same embedding techniques are the conceptual foundation for the document embeddings used in modern retrieval systems, including retrieval-augmented generation, where entire chunks of text, rather than single words, are converted into vectors so that semantically similar pieces of text can be found through vector similarity search.

The Transformer Revolution in NLP

The introduction of the transformer architecture in 2017 fundamentally changed the trajectory of NLP. Transformers use a mechanism called self-attention to weigh the relevance of every word in a sentence to every other word simultaneously, which allows the model to capture long-range dependencies and contextual meaning far more effectively than earlier recurrent architectures. This led to the development of large pretrained language models, such as BERT, introduced in 2018 for understanding tasks, and the GPT family of models, designed for text generation. These models are first pretrained on massive amounts of unlabeled text using a generic objective, such as predicting a missing or next word, and are then adapted, or fine-tuned, to specific downstream tasks with comparatively small amounts of labeled data. This pretrain-then-adapt paradigm, sometimes extended through techniques like prompting rather than fine-tuning, is now the dominant approach across nearly all of modern NLP, and is the technology underlying contemporary conversational AI systems and retrieval-augmented generation applications like document question-answering bots.

Evaluating NLP Systems

Because language tasks often have more than one acceptable correct answer, evaluating NLP systems is inherently more complex than evaluating a simple classifier. Metrics such as BLEU

and ROUGE are commonly used to automatically score machine translation and summarization systems by comparing generated text to one or more human-written reference texts, although these metrics are known to correlate imperfectly with human judgments of quality. For tasks like classification, sentiment analysis, or named entity recognition, standard machine learning metrics such as accuracy, precision, recall, and F1 score remain the norm. For generative and conversational systems, human evaluation, in which people directly rate the quality, relevance, and correctness of generated responses, remains an important complement to automatic metrics.

NLP and Retrieval-Augmented Generation

Retrieval-augmented generation, the technique underlying this document question-answering system, is itself a direct application of core NLP ideas. It combines a retrieval step, which uses text embeddings and vector similarity search, both concepts rooted in the word and document embedding techniques described above, with a generation step, which uses a large pretrained language model of the kind described in the transformer section, conditioned on the retrieved text to produce a grounded, human-readable answer. This combination allows a language model to answer questions about specific documents it was never trained on, without needing to be retrained, simply by supplying the relevant text as additional context at the time the question is asked.